Inflationary Cosmology

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1 Introduction

In this course, we have covered much of the current theory of the early universe. One of the key underlying questions motivating the study is how our universe came to be with cosmic structures arranged in their particular way. Many topics in the course stem from this, including baryogenesis, different eras of the universe (in particular, a radiation-dominated one and a matter-dominated one), and the development of the cosmic web resulting from quantum fluctuations.

However, all of these subjects stem from the origins of structure in the universe, and the current size of the universe as compared to theory permits only a very short time for the universe to "inflate" to a certain distance, hence the notion of cosmic inflation. This theory posits the idea that the universe has scale factor a (determining the relative scale between two points as they were compared to now) that had grown by an exponential amount.

In this paper, we will discuss the theory motivating cosmic inflation and the consequences of quantum fluctuations as creators of structure. Once we lay the groundwork, we will integrate the theory of cosmic inflation to the classical model, through the slow-roll inflation model. This will provide us ways to explain certain experimental facts in the early universe, such as the CMB, the homogeneity problem, and the flatness problem. Of course, there is always more progress to be made on the theory, and we will point out certain theorized phenomena from cosmic inflation that have yet to be confirmed as well as some further theories about our universe building on it (e.g. ekpyrotic models and eternal chaotic inflation).

2 Single Scalar Fields in the Framework of Inflation

In this section, we will go through the major equations used in deriving the exponential speed of inflation of the scale factor using the Friedmann equations, following the Einstein Field Equations. This will help recap some of the results from the course lecture notes which can then extend to further developments within the theory.

2.1 Definition of the Lagrangian and the Metric

The Lagrangian for scalar field inflation including gravity is

$$\mathcal{L} = -\frac{1}{2}\sqrt{-g}[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi)].$$

We will be using the Friedmann-Lemaitre-Robertson-Walker metric (FLRW metric) to allow for a universe evolving over space-time. The metric in standard rectangular coordinates is of the form

$$ds^2 = -dt^2 + a^2(t)(dx^2 + dy^2 + dz^2),$$

such that $a(t)$ is the scale factor of space-time at time t. In particular, we require $a(t_{\text{today}}) = 1$. We prefer to work in conformal time in order to group all of the variables in the metric under a single scale factor. Define

$$d\eta = \frac{dt}{a}$$

so that we get

$$ds^2 = a^2(t)(-d\eta^2 + dx^2 + dy^2 + dz^2)$$

as the FLRW metric in conformal time coordinates.

2.2 The Friedmann Equation and the Speed of Inflation

Using the Einstein Field Equation, that is

$$G^{\mu\nu} + \Lambda g^{\mu\nu} = 8\pi G T^{\mu\nu}$$

and the FLRW metric above, under, and the assumption that the universe evolves as a perfect fluid, we get the **Friedmann Equation**

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3} - \frac{k}{a^2} + \frac{\Lambda}{3}$$

where ρ is the energy density, k is the curvature, and H is the Hubble constant, defined as

$$H = \frac{\dot{a}}{a}$$

In the time of inflation, the first term was the dominant term with constant energy density ρ_0, dictated by the potential of the scalar field $V(\phi)$ so that

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho_0}{3}$$

which is to say H is constant in time during this period. This leaves a rather nice ODE

$$H = \frac{\dot{a}}{a}.$$

Solving this with initial condition $a(0) = a_0$ yields

$$a(t) = a_0 e^{Ht}.$$

and thus we see the exponential scaling of a in the period of cosmic inflation.

3 Slow-Roll Inflation and Quantum Fluctuations

As of right now, we have determined a model that can describe the exponential expansion of the universe, but there is a key physical constraint to this model. The theory as described in section 2 allows no current way to transition from inflation to the reheating period of the universe when the evolving universe slows down in a to create the universe we live in. The solution to this "graceful exit problem" gives the motivation for slow-roll inflation [Lin90]. Hubble friction was introduced, a gravitational mechanism that sustains inflation for a short enough period that is reasonable from current experimental theory. This leads to the slow-roll inflation model, and

coupled with quantum fluctuations gives a rather intuitive origin for the structures that we can observe in the modern universe (e.g. galaxy clusters, stars, planets, etc).

3.1 Description of the Slow-Roll Model

We capture the evolution of inflation and its transition to the current scaling of a via the following diagram:

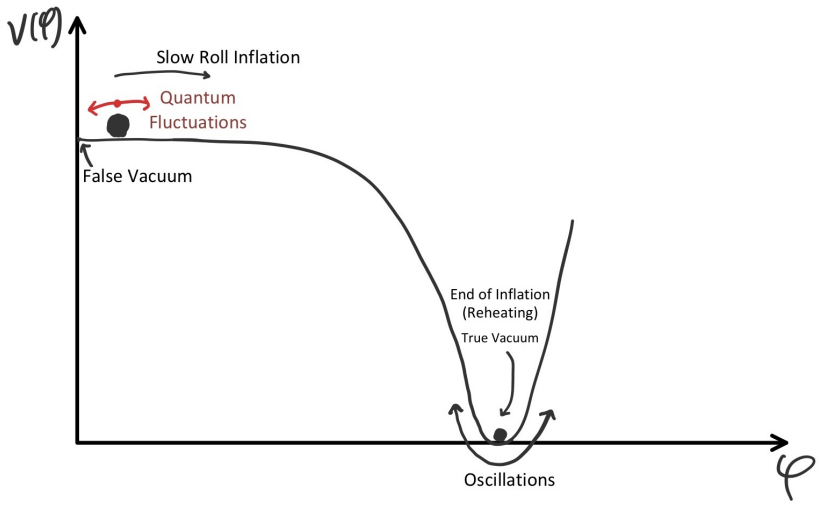


Figure 1: The potential for slow-roll inflation with the field beginning at a false vacuum.

First, the inflation begins at a false vacuum state, which is the metastable point that is not an actual minimum of the potential. Quantum fluctuations ensure that any deviation from the metastable extrema ends up having a "lemming," as Professor Susskind puts it, roll down the potential at a slow rate. Once fallen into the well, the "lemming" begins to oscillate around a true minima of the potential, and this oscillation is the process that ends the inflationary period.

3.2 Slow-Roll Inflation

We adopt the material and conventions in [Flo25]. First, we consider the Lagrangian from section 2.1 but in flat-space coordinates, so that

$$S = \int -\frac{1}{2}[\partial_\mu \phi \partial^\mu \phi] - V(\phi).$$

In this case, the Euler-Lagrange equations become

$$\ddot{\phi} - \nabla^2 \phi = -\frac{\partial V}{\partial \phi}$$

where converting this Klein-Gordon equation of motion to one created by the FLRW metric yields

$$\ddot{\phi} + 3H\dot{\phi} = -\frac{\partial V}{\partial \phi}.$$

We now wish to define energy density under this Lagrangian, setting

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$

with the continuity equation such that

$$\dot{\rho}_\phi = -3H(\rho_\phi + P_\phi).$$

Combining with above equations, we get that the pressure is

$$P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi)$$

which gives an equation of state

$$w_\phi = \frac{\rho_\phi}{P_\phi}$$

which for when the potential term dominates, we have that $w_\phi \approx -1$. This makes the Hubble constant approximately constant so that we get an exponential growth in a overtime. We will focus on this assumption to determine the period of inflation. We now proceed with the slow-roll assumption. In this model, we take take the scalar field as the dominant term for energy density so that

$$H^2 = \frac{1}{3M_{pl}^2}\left[\frac{1}{2}\dot{\phi}^2 + V(\phi)\right]$$

where M_{pl} is the reduced Planck mass. We now take the slow-roll parameters

$$\varepsilon = -\frac{\dot{H}}{H},\ \kappa = \frac{\dot{\epsilon}}{H\epsilon}$$

and work in the approximation where the potential term dominates.

$$H^2 = \frac{V}{3M_{pl}^2}$$

with approximated Klein-Gordon equation

$$3H\dot{\phi} = -\frac{\partial V}{\partial \phi}$$

Due to the exponential nature of a, we can let $N = \ln a$, the number of e-folds during inflation. We get that

$$N = \int_{t_i}^{t_e} H(t)\ dt = \int_{\phi_i}^{\phi_e} \frac{H}{\dot{\phi}} d\phi$$

so that $\varepsilon(t_i) = \varepsilon(t_f) = 1$. Given this value, we can determine that $N \approx 60$ as stated in class!

3.3 Cosmological Perturbation Theory

To actually work with quantum fluctuations, we first need to develop some perturbation theory. We will adopt material from [Bau11] [Flo25]. We consider perturbations to the metric and energy density

$$g_{\mu\nu} \mapsto g_{\mu\nu} + \delta g_{\mu\nu}$$

$$\rho \mapsto \overline{\rho} + \delta\rho.$$

As a note, such perturbations can depend on position, which we would want in this case to motivate quantum fluctuations and the small asymmetries in the universe. We rewrite the FLRW metric in conformal time with perturbations of the form

$$ds^2 = a^2(\eta) = [-(1+2\Psi)d\eta^2 + [1-2\Phi]dx^j dx^j]$$

where we work in the Newtonian gauge and $\Psi = \Phi$, which the naming follows from the fact that both Ψ and Φ are acting as some sort of gravitational potential within the metric. As we vary the action induced by the Lagrangian for a single-particle scalar for an arbitrary metric, we get the **Mukhanov-Sasaki Equation**

$$v_k'' + \left(k^2 - \frac{z''}{z}\right) v_k = 0$$

such that $v_k = a(\delta\phi)$ and $z = \frac{\overline{\phi}}{H}$. Now, with slow-roll inflation, we have H constant and $a = -(H\eta)^{-1}$ so that the differential equation becomes

$$v_k'' + \left(k^2 - \frac{2}{\eta^2}\right) v_k = 0.$$

3.4 Quantum Fluctuations and the Scalar Spectral Index

We first canonically quantize v_k to build the theory on quantum fluctuations. Writing v_k in position space as $\hat{v}(\eta, x)$ using a Fourier Transform and its conjugate momentum as $\hat{\pi}(\eta, x)$, we impose the commutation relation

$$[\hat{v}(\eta, x), \hat{\pi}(\eta, x')] = i\delta^{(3)}(x - x')$$

and using ladder operators (just like in the quantization of the harmonic oscillator), we rewrite

$$\hat{v}_k(\eta) = v_k(\eta)\hat{a}_k + v_k^*\hat{a}_{-k}^\dagger.$$

The goal of this is to find a vacuum state so that

$$\hat{a}_k(\eta)\,|0\rangle = 0$$

and in the limit where $|k\eta| >> 1$ (i.e. near the end of the inflationary period), we take modes of the form

$$\omega_k^2 = k^2 - \frac{2}{\eta^2} = k^2\left(1 - \frac{2}{(k\eta)^2}\right) \approx k^2$$

which makes the Mukhanov-Sasaki Equation just an ODE with solutions of complex exponential modes. Thus, we can force an initial condition

$$\lim_{k\eta\to\infty} v_k(\eta) = \frac{1}{\sqrt{2k}} e^{-ik\eta}$$

which, when solving the full differential equation, gives us

$$v_k(\eta) = \frac{1}{\sqrt{2k}} \left(1 - \frac{i}{k\eta}\right) e^{-ik\eta}.$$

This equation is known as the **Bunch-Davies mode function**. The corresponding vacuum state associated with this function is known as the **Bunch-Davies vacuum**. With this, some computations with integrals and commutators will yield

$$\langle |\hat{v}|^2 \rangle = \int d\ln k \frac{k^3}{2\pi^2} |v_k(\eta)|^2.$$

The integrand allows us to naturally define a power spectrum density of the form

$$\Delta_v^2(k,\eta) = \frac{k^3}{2\pi^2} |v_k(\eta)|^2$$

and using $v = a(\delta\phi)$, we have that

$$\Delta^2_{\delta\phi}(k,\eta) = \frac{\Delta_v^2(k,\eta)}{a^2(\eta)} = \frac{k^3}{4k\pi^2 a^2(\eta)} \left(1 + \frac{1}{(k\eta)^2}\right) = \frac{1}{4\pi^2 a^2(\eta)\eta^2}(1 + (k\eta)^2) = \left(\frac{H}{2\pi}\right)^2 (1 + (k\eta)^2)$$

in the limit where $|k\eta| \to \infty$ as before when we exit inflation, we see that

$$\Delta^2_{\delta\phi}(k,\eta) \to \left(\frac{H}{2\pi}\right)^2$$

which shows that this power spectrum is scaleless over modes! We now use this to analyze curvature perturbations to our metric, defining $\mathcal{R} = \frac{aH}{\dot{\phi}'}\delta\phi$ as our curvature perturbation so that (using ε from one of our slow-roll parameters)

$$\Delta^2_{\mathcal{R}} = \left(\frac{H\Delta_{\delta\phi}}{\dot{\phi}}\right)^2 = \frac{1}{8\pi^2\varepsilon}\left(\frac{H}{M_{pl}}\right)^2.$$

Lastly, define the **scalar spectral index** n_s as

$$n_s - 1 = \frac{d\ln\Delta^2_{\mathcal{R}}}{d\ln k}.$$

Using both slow-roll parameters ε and κ, we have that by using the above two equations shows that

$$n_s - 1 = -2\varepsilon - \kappa$$

which will become helpful to analyze next section using experimental evidence from the CMB.

4 Experimental Evidence and the Strength in the Theory

As the theory of cosmic inflation has been discussed, we will review ideas that were considered major problems in the theory of the early universe, including the homogeneity problem, the

flatness problem, and the problem of magnetic monopoles. Through inflation, such problems can be realized with more reasonable assumptions. Through some experimental results coming from the CMB, we will investigate how inflation matches the results of the CMB. Because of these density perturbations, however, Guth proposes a couple of questions that result from this.

4.1 The Homogeneity Problem

Guth explains the homogeneity problem (also known as the horizon problem) as the fact that the universe is rather homogeneous with fluctuations in density rather low as seen through the CMB [Gut81]. Nevertheless, there are many different regions in our universe that are causally disconnected by the speed of light, so information could not possibly have been communciated between the vast majority of space-time. Guth speculates that a solution to this problem must imply the existence of a causally-violating force in the early period of the universe, and the inflationary model is just the solution. As the early universe was assumed to be confined to a space that is causally connected, such a problem would naturally resolve itself by the approximate homogeneity created during the big bang, preserved through the eventual inflation of the universe.

4.2 The Flatness Problem

Guth's proposal of cosmic inflation handles the original problem of approximate flatness as well in [Gut81]. Define

$$\Omega = \frac{\rho}{\rho_c} = \frac{8\pi G\rho}{3H^2}$$

where ρ_c, the critical, energy density of the universe if it was spatially flat and ρ the current energy density. Note that with $\Omega = 1$ being an unstable equilibrium, any deviation from it would be amplified in time. In particular, there is a notion of needing to "fine-tune" H_0 given the initial energy scale T_0. Given $T_0 = 10^{17}$ GeV, one would require a fine-tuning on the order of 10^{55} without any theory of (i.e. with only classical cosmology). Guth further notes that

$$\Omega - 1 = \begin{cases} t \text{ matter dominated universe} \\ t^{2/3} \text{ radiation dominated universe} \end{cases}$$

which, notably are both drastically slower in growth than that of exponential type [Gut07]. In particular, calculations show that going to the Planck second after the big bang, fine-tuning Ω would require it to be one to one part in 59. However, the theory of cosmic inflation allows for

$$\Omega - 1 \propto e^{-2Ht}$$

which although given the small amount of time for inflation, allows for a much larger deviation for Ω from one than the current theory of classical cosmology.

4.3 The Problem with Magnetic Monopoles

Guth emphasizes that without cosmic inflation, it can be shown that classical cosmology predicts the existence of magnetic monopoles, or particles with their own magnetic charge [Gut07]. In fact, such a model would have such magnetic monopoles produced at the order of 10^{12} more than all other matter in the universe, which is certainly not known to be experimentally true at all [Pre79]. With cosmic inflation, as long as the magnetic monopole production ended before the end of inflation, the monopole density would decrease almost immediately. This would explain the fact that we are not able to observe any magnetic monopoles in the known visible universe.

4.4 Verification through the CMB

Recall n_s as the scalar spectral index which depends on k, the momentum, characterizing the intensity of the fluctuations in the temperature of the CMB [Gut13]. In particular, we have that using slow-roll parameters from section 3.2, we get that

$$n_s - 1 = -2\varepsilon - \kappa \approx 4\frac{\dot{H}}{H^2} - \frac{\ddot{H}}{H\dot{H}}$$

where experimental data gives that $n_s - 1 = -0.032 \pm 0.012$. This rather small value becomes important due to the fact that $n_s = 1$ implies a scale-invariance across all modes. This matches with the idea of single-field inflation since density fluctuations produced at the time are scale invariant with most modes generated under approximately a similar period of time in inflation, allowing them all to be roughly similar. Furthermore, experimental tests with the CMB also confirm that Ω_Λ, Ω_{baryon}, and $\Omega_{\text{dark matter}}$ all agree with values from $n_s \approx 1$.

4.5 Questions on Density Perturbations

As a result of the density fluctuations in the temperature pattern of the CMB, there are some questions that seem to provide more insight based on the assumption of quantum fluctuations. Guth [Gut13] proposes :

Question 1: Do there exist B-modes? Focusing on the polarization of the CMB, there has been an experimental focus on using spherical harmonic expansions of the CMB to isolate E-modes and B-modes using the temperature of the CMB. Let E-modes be those that can be expressed as the gradient of a scalar harmonic function. B-modes are those orthogonal to E-modes, and it seems that density perturbations, while being able to create E-modes, do not seem to be able to generate B-modes. The current known source of primordial B-modes originate from primordial gravitational waves, but many inflationary theories have energy scales that are lower than those that can produce such gravitational waves. The current energy scale limits of cosmic inflation originate anywhere from the Electroweak to the GUT scale, however.

Question 2: Can we use cosmic inflation to understand Sub-planckian physics? Recall that during the period of inflation, the scale factor grows by 60 e-fold, which is approximately 10^{26}. After the inflationary period, the scale factor up to now has grown an additional factor of 10^{27}, implying that lengths that are currently around 1 meter were scaled from 10^{-53} meters at the beginning of inflation. This is far smaller than the Planck scale. One theory proposed by [Kal+02] posits that perturbations are a result of effective field theories, making

any sub-Planckian effects essentially irrelevant as space-time evolves. Nevertheless, it is still up to speculation.

Question 3: Should there be further improvements to the current slow-roll inflation model? As of right now, the current model explained earlier in this section results in a rather simple description of a potential driving slow-roll inflation with only a single scalar field Lagrangian. The Wilkonsin Microwave Anistropy Probe, or WMAP, has been used as experimental evidence to justify the current model's simplicity, but there may very well be more complications in the model as future experimental data provides results that are not able to originate from the simple model.

5 Alternatives and Further Hypothesis for Cosmic Inflation

Of course, there is always a question of what drove the inflation as well as some theoretical consequences of this theory. In this section, we will highlight the key theories that can prescribe alternative models and further hypothesis that delve into some of the strange results that emerge from this field.

5.1 Ekpyrotic or Cyclic Model

Proposed by Steinhardt and Turok, this inflationary model assumes no "beginning of time," but rather an infinite repeating cycle of inflation and contractions [ST02]. The model starts off with the current known "standard model" of inflation and post-inflation period, but it is further proposed that the scalar field ϕ will "roll back" to $-\infty$ and cause a contraction of a universe. Reducing the entropy during this contraction is believed to be a result of dark energy in this stage of the unvierse. To summarize the model as a whole, the scale factor of a increases as the universe inflates through its different epochs (e.g. matter-dominated and radiation-dominated), and the roll back will have the kinetic energy term dominate the Lagrangian again (as opposed to potential energy during inflation).

5.2 Eternal Inflation

Steinhardt and Vilenkin originally posited this theory. Guth explains that the decay of a false vacuum state does not necessarily just imply the existence of a single universe [Gut07]. In particular, the false vacuum state, although it decays, grows exponentially even faster than it decays. Thus, the false vacuum state remains, and due to the exponential growth of the false vacuum state, eventually reaches the original size of the false vacuum, leading to a new decay in each region. This in fact leads to a new set of inflations for universes disjoint from the original, and this process repeats arbitrarily long. Each new universe created is called a bubble universe, and are created at an exponential rate. Although this is not experimentally verifiable (at least for now), this has been theorized to be an immediate result of the current framework of inflation.

5.3 Eternal Chaotic Inflation

Proposed by Andrei Linde, chaotic inflation suggests that the inflation occurs not starting at a plateau like before, but rather already beginning on a downhill slope, around a potential that looks approximately quadratic [Lin86].

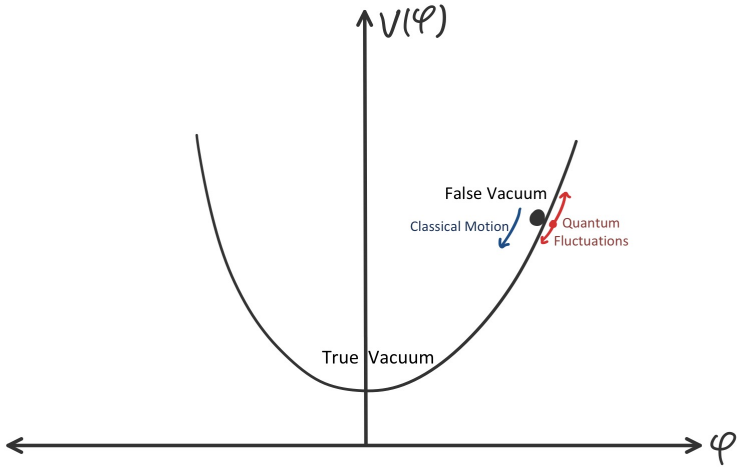


Figure 2: The potential for chaotic inflation with the field beginning in a false vacuum.

Linde showed that this case of inflation can also be eternal, hence the term eternal chaotic inflation. Guth builds off of this, explaining that the motion of the field created by this potential would be affected not only by any classical motion but also any quantum fluctuations as well [Gut07]. In one Hubble time (that is, H^{-1}), Hubble volume (H^3) expands by a factor of e^3, or about 20 times. We consider that the change in the field can be written as

$$\Delta\phi = \Delta\phi_{\text{classical}} + \Delta\phi_{\text{quantum}}$$

of which the two terms on the right hand side have a probability of $\frac{1}{20}$ for being both positive. As long as there is a probability larger than $\frac{1}{20}$ of the field increasing the Hubble volume in Hubble time, then eternal inflation will occur. This is simply a result of the fact that as long as the number of inflated regions is guaranteed to increase, then the logic from Section 5.2 permits further inflations to occur.

6 Conclusion

We have gone through many of the highlights related to cosmic inflation throughout this paper. First, we assumed a Lagrangian of the form taken for a single-particle scalar field (under any arbitrary metric). Using the FLRW metric as well, we are able to derive the speed of inflation for a, increasing at an exponential rate. We built on the theory of inflation to integrate a "slow-roll model" that allows a smooth transition to the reheating phase of the universe. The initial drive of the slow-roll model comes from quantum fluctuations which under the formulation of cosmological perturbation theory allows one to determine theoretically the power spectrum density and the scalar spectral index. With the theory established, many problems in cosmology have fit into the theory of inflation, including the reason for almost perfect homogeneity across the universe, an energy density that emerges from one very close to that coming from a spatially flat universe, and the lack of magnetic monopoles in the present day. We verify the results related to slow-roll inflation via the CMB temperature fluctuations, and see that the data aligns with predicted values, helping to validate our theory of inflation. All of this being said, there

are many consequences of this theory that provide more questions and hypotheses, such as the ekpyrotic or cyclic models of the universe, and an eternal chaotic inflation.

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